C.3. Real-world Challenges in Pest Detection using Deep Learning: an Investigation into Failures and Solutions

C.3.1. THE AI SCIENTIST-V2 Idea

Idea

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"Name": "real_world_pest_detection",
"Title": "Real-World Challenges in Pest Detection Using Deep Learning: An Investigation into
Failures and Solutions",
"Short Hypothesis": "Deep learning models for pest detection often fail to generalize in real-
world agricultural settings due to data quality issues, environmental variability, and model
limitations. Investigating these failures can lead to more robust solutions.",
"Related Work": "Several studies, such as those by Agarwal et al. (2023) and Dong et al.
(2024), have explored deep learning for pest detection in agriculture. These studies generally
report high accuracy in controlled settings but often do not address real-world deployment
challenges. Our proposal distinguishes itself by focusing on the negative outcomes and the
underlying reasons behind these failures.",
"Abstract": "Accurate pest detection is vital for protecting crops and ensuring food security.
While deep learning models have shown promise in controlled environments, their performance
often degrades in real-world applications. This proposal aims to investigate the reasons behind
these failures. We hypothesize that data quality issues, environmental variability, and model
limitations are significant factors. By conducting a series of experiments, we will explore
these challenges in depth and propose robust solutions to improve the generalizability of deep
learning models for pest detection. Our research will provide valuable insights for the
agricultural community and contribute to the development of more reliable AI tools for
precision farming.",
"Experiments": [
    "1. **Data Quality Analysis**: Collect a diverse dataset of pest images from different
    agricultural environments and analyze its quality. Identify common issues such as label
    noise, class imbalance, and distribution shift.",
    "2. **Model Robustness Testing**: Train state-of-the-art deep learning models (e.g.,
    YOLOv8, EfficientNetB3) on the collected dataset and evaluate their performance in
    controlled vs. real-world settings. Metrics: Mean Average Precision (mAP), F1 Score.",
    "3. **Environmental Variability Study**: Evaluate model performance under different
    environmental conditions (e.g., lighting, weather). Identify which conditions most
    significantly impact model accuracy.",
    "4. **Failure Mode Analysis**: Conduct a detailed analysis of misclassifications to
    identify common patterns and potential causes (e.g., feature overlap between pests and
    background).",
    "5. **Improvement Strategies**: Implement and test various strategies to mitigate
    identified challenges, such as data augmentation, domain adaptation, and model ensembling.
    Evaluate their effectiveness in improving model robustness."
],
"Risk Factors and Limitations": "Potential risks include the availability and quality of real-
world data, the computational demands of training and testing multiple deep learning models,
and the generalizability of the findings to different types of pests and crops. Additionally,
environmental factors may introduce variability that is challenging to control.",
"Code": "from datasets import ..."
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Link to more material: https://github.com/SakanaAI/AI-Scientist-ICLR2025-Workshop-Experiment/tree/master/pest-detection.

Note on dataset preparation: As mentioned in section 4.2, we selected three AI-generated ideas–two from the core machine learning research batch and one from the more applied, real-world-oriented batch. This pest detection idea comes from the latter. While the system performs well when downloading standard machine learning datasets, it still struggles to automatically access real-world datasets available online. To address this, we manually downloaded a relevant dataset from Kaggle (Crop Pest and Disease Detection[1]) and reduced its size to one tenth to speed up our experiments. Automating this data preparation process further to handle a broader range of datasets remains an interesting direction for future research.

[1] https://www.kaggle.com/datasets/nirmalsankalana/crop-pest-and-disease-detection